

tf.constant

 [View source on GitHub \(https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/framework/constant_op.py#L177-L277\)](https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/framework/constant_op.py#L177-L277)

Creates a constant tensor from a tensor-like object.

```
tf.constant(  
    value, dtype=None, shape=None, name='Const'  
) -> Union[tf.Operation (https://www.tensorflow.org/api\_docs/python/tf/Operation), ops._EagerTensorBase]
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Better performance with tf.function (https://www.tensorflow.org/guide/function) • TF-NumPy Type Promotion (https://www.tensorflow.org/guide/tf_numpy_type_promotion) • Introduction to Tensors (https://www.tensorflow.org/guide/tensor) • Introduction to graphs and tf.function (https://www.tensorflow.org/guide/intro_to_graphs) • Migrate `tf.feature_column`'s to Keras preprocessing layers (https://www.tensorflow.org/guide/migrate/migrating_feature_columns)	• DeepDream (https://www.tensorflow.org/tutorials/generative/deepdream) • Load text (https://www.tensorflow.org/tutorials/load_data/text) • Playing CartPole with the Actor-Critic method (https://www.tensorflow.org/tutorials/reinforcement_learning/actor_critic) • Neural style transfer (https://www.tensorflow.org/tutorials/generative/style_transfer) • Custom training loop with Keras and MultiWorkerMirroredStrategy (https://www.tensorflow.org/tutorials/distribute/multi_worker_with_ctl)

★ **Note:** All eager [tf.Tensor](https://www.tensorflow.org/api_docs/python/tf/Tensor) (https://www.tensorflow.org/api_docs/python/tf/Tensor) values are immutable (in contrast to [tf.Variable](https://www.tensorflow.org/api_docs/python/tf/Variable) (https://www.tensorflow.org/api_docs/python/tf/Variable)). There is nothing especially *constant* about the value returned from [tf.constant](https://www.tensorflow.org/api_docs/python/tf/constant) (https://www.tensorflow.org/api_docs/python/tf/constant). This function is not fundamentally different from [tf.convert_to_tensor](https://www.tensorflow.org/api_docs/python/tf/convert_to_tensor) (https://www.tensorflow.org/api_docs/python/tf/convert_to_tensor). The name [tf.constant](https://www.tensorflow.org/api_docs/python/tf/constant) (https://www.tensorflow.org/api_docs/python/tf/constant) comes from the **value** being embedded in a **Const** node in the [tf.Graph](https://www.tensorflow.org/api_docs/python/tf/Graph) (https://www.tensorflow.org/api_docs/python/tf/Graph). [tf.constant](https://www.tensorflow.org/api_docs/python/tf/constant) (https://www.tensorflow.org/api_docs/python/tf/constant) is useful for asserting that the value can be embedded that way.

If the argument **dtype** is not specified, then the type is inferred from the type of **value**.

```
>>> # Constant 1-D Tensor from a python list.  
>>> tf.constant([1, 2, 3, 4, 5, 6])  
<tf.Tensor: shape=(6,), dtype=int32, numpy=array([1, 2, 3, 4, 5, 6], dtype=int32)>  
>>> # Or a numpy array  
>>> a = np.array([[1, 2, 3], [4, 5, 6]])  
>>> tf.constant(a)  
<tf.Tensor: shape=(2, 3), dtype=int64, numpy=array([[1, 2, 3], [4, 5, 6]])>
```

If **dtype** is specified, the resulting tensor values are cast to the requested **dtype**.

```
>>> tf.constant([1, 2, 3, 4, 5, 6], dtype=tf.float64)  
<tf.Tensor: shape=(6,), dtype=float64, numpy=array([1., 2., 3., 4., 5., 6.])>
```

If **shape** is set, the **value** is reshaped to match. Scalars are expanded to fill the **shape**:

```
>>> tf.constant(0, shape=(2, 3))  
<tf.Tensor: shape=(2, 3), dtype=int32, numpy=
```

```

array([[0, 0, 0],
       [0, 0, 0]], dtype=int32)>
>>> tf.constant([1, 2, 3, 4, 5, 6], shape=[2, 3])
<tf.Tensor: shape=(2, 3), dtype=int32, numpy=
  array([[1, 2, 3],
         [4, 5, 6]], dtype=int32)>

```

`tf.constant` (https://www.tensorflow.org/api_docs/python/tf/constant) has no effect if an eager Tensor is passed as the `value`, it even transmits gradients:

```

>>> v = tf.Variable([0.0])
>>> with tf.GradientTape() as g:
...     loss = tf.constant(v + v)
>>> g.gradient(loss, v).numpy()
array([2.], dtype=float32)

```

But, since **`tf.constant`** (https://www.tensorflow.org/api_docs/python/tf/constant) embeds the value in the **`tf.Graph`** (https://www.tensorflow.org/api_docs/python/tf/Graph) this fails for symbolic tensors:

```

>>> with tf.compat.v1.Graph().as_default():
...     i = tf.compat.v1.placeholder(shape=[None, None], dtype=tf.float32)
...     t = tf.constant(i)
Traceback (most recent call last):
...
TypeError: ...

```

`tf.constant` (https://www.tensorflow.org/api_docs/python/tf/constant) will create tensors on the current device. Inputs which are already tensors maintain their placements unchanged.

Related Ops:

- **`tf.convert_to_tensor`** (https://www.tensorflow.org/api_docs/python/tf/convert_to_tensor) is similar but:
 - It has no `shape` argument.
 - Symbolic tensors are allowed to pass through.

```

>>> with tf.compat.v1.Graph().as_default():
...     i = tf.compat.v1.placeholder(shape=[None, None], dtype=tf.float32)
...     t = tf.convert_to_tensor(i)

```

- **`tf.fill`** (https://www.tensorflow.org/api_docs/python/tf/fill): differs in a few ways:
 - **`tf.constant`** (https://www.tensorflow.org/api_docs/python/tf/constant) supports arbitrary constants, not just uniform scalar Tensors like **`tf.fill`** (https://www.tensorflow.org/api_docs/python/tf/fill).
 - **`tf.fill`** (https://www.tensorflow.org/api_docs/python/tf/fill) creates an Op in the graph that is expanded at runtime, so it can efficiently represent large tensors.
 - Since **`tf.fill`** (https://www.tensorflow.org/api_docs/python/tf/fill) does not embed the value, it can produce dynamically sized outputs.

Args	
value	A constant value (or list) of output type dtype .
dtype	The type of the elements of the resulting tensor.
shape	Optional dimensions of resulting tensor.
name	Optional name for the tensor.

Returns	
A Constant Tensor.	
Raises	
TypeError	if shape is incorrectly specified or unsupported.
ValueError	if called on a symbolic tensor.

Except as otherwise noted, the content of this page is licensed under the [Creative Commons Attribution 4.0 License](https://creativecommons.org/licenses/by/4.0/) (<https://creativecommons.org/licenses/by/4.0/>), and code samples are licensed under the [Apache 2.0 License](https://www.apache.org/licenses/LICENSE-2.0) (<https://www.apache.org/licenses/LICENSE-2.0>). For details, see the [Google Developers Site Policies](https://developers.google.com/site-policies) (<https://developers.google.com/site-policies>). Java is a registered trademark of Oracle and/or its affiliates. Some content is licensed under the [numpy license](https://numpy.org/doc/stable/license.html) (<https://numpy.org/doc/stable/license.html>).

Last updated 2024-04-26 UTC.